

Activity trace — extending matched SHD training

ar071 · 19 July 2026 · Draft

Checkpoint CP-001

This publication-to-mandate checkpoint has immutable private source SHA-256 prefix `4b08b55d459f`. Its checkpoint time is `2026-07-19 00:49:00.457 UTC`. It has no predecessor within this night. Hidden reasoning, tool payloads, injected environment material, credentials, private paths, and infrastructure identifiers are excluded; the one visible private local link is explicitly redacted below.

Decision and action ledger

1. Pull request 51 was explicitly accepted and merged into `main` using merge commit `569926f`; no squash or rebase was used.
2. The accepted result motivates higher accuracy, but repeated official-test use was rejected as an optimization loop. The next outcome is held-out validation accuracy on `exp068`'s identical deterministic split.
3. The approved intervention changes only fixed training duration, from 40 to 80 epochs, in both matched cells. One seed remains appropriate for this exploratory mechanism check.
4. New identifiers are `ar070`, `ar071`, and `exp069`. At approval, `exp069` spend was 0 USD, no pod was active, and no implementation or result existed.

Privacy-sanitized visible transcript

Assistant

2026-07-19 00:41:04.944 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`Exp068` is complete and stopped at the human-review gate. `PR 51` remains draft and unmerged.

Key evidence:

- PING: **45.72%** official-test accuracy
- COBA: **35.07%**
- Difference: **+10.64 percentage points**
- PING/COBA macro accuracy: 45.47% / 34.64%
- PING/COBA cross-entropy: 2.055 / 2.345
- PING E/I activity: 10.50 / 47.99 Hz
- COBA E activity: 28.44 Hz
- One seed; all 2,264 official-test samples evaluated after paired checkpoint freeze
- Zero skipped updates or non-finite batches
- Exact recorded compute spend: **2.4414148 USD**
- Active pods: **0**

Links included PR 51 and the rendered ar068, exp068, and ar069 pages. (PRIVATE LOCAL PATH REDACTED)

Validation passed: focused Ruff and ty, compilation, 20 artifact hashes, registered-result assertions, raster inspection, zero-stub 35-entry Demolab build, and GitHub preview. Repository-wide CI lint remains red solely from previously documented ty debt; the test job was still running at handoff.

Final commits: 7671ce4 and ae7a26a. Goal usage: 5,630 tokens over roughly 18 seconds of tracked goal execution.

User

2026-07-19 00:44:59.911 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

merge into main, then make a new pr we gonna try get those acc numbers higher

Assistant

2026-07-19 00:45:05.441 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll merge PR 51 into `main` at the publication gate first, preserving its merge history. Then I'll synchronize `main` and open a fresh research branch/PR for the accuracy-improvement night; I'll keep that new experiment at the mandate-design gate until its scientific setup is explicit. I'm using the GitHub workflow guidance because this changes PR and branch state.

Assistant

2026-07-19 00:45:15.365 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is mergeable and the preview is green. Its lint failure is still only the previously documented repository-wide type debt, with no exp068 diagnostic; the long test job remains in progress. Your explicit merge decision resolves the publication gate, so I'm marking the draft ready and merging it now without squashing.

Assistant

2026-07-19 00:45:28.680 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit 569926f. This closes the exp068 publication gate. I'm synchronizing that accepted state and checking the autoresearch registration rules and next available IDs before creating the new accuracy-improvement PR.

Assistant

2026-07-19 00:46:05.090 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit `569926f`.

Before opening the next PR, the autoresearch mandate gate requires a locked scientific design. I recommend this clean next step:

- One seed, same deterministic 7,340/816 SHD development split.
- Keep the official test sealed; repeated use would turn it into an optimization target.
- Train matched COBA and PING for 80 epochs instead of 40.
- Preserve every `exp068` architecture, optimizer, input, and readout setting.
- Select checkpoints using validation accuracy, then cross-entropy, then earlier epoch.
- Primary criterion: PING improves at least 3 percentage points over its `exp068` 43.50% validation baseline.
- Secondary: whether COBA improves over 40.81%, whether PING remains ahead, and activity/runtime diagnostics.
- One local smoke, then two concurrent pods, 10 USD ceiling, no `tools/snn` edits.
- New IDs: `ar070`, `ar071`, and `exp069`.
- Only after approval: lock the mandate on `main`, create the night branch, and open the new draft PR.

Reply `go` to approve this mandate.

User

2026-07-19 00:48:51.919 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`go`

Assistant

2026-07-19 00:49:00.457 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Mandate approved. I'm locking `ar070`, `ar071`, and `exp069` on `main` now with the 80-epoch, validation-only exploratory contract. This approval starts no compute: after the anchor is pushed, I'll create the dedicated night branch and new draft PR from that exact commit.

Checkpoint CP-002

This branch-and-PR checkpoint follows CP-001. Its immutable private source has SHA-256 prefix `77b86b9fea8d`, and its checkpoint time is `2026-07-19 00:53:19.217 UTC`.

Action ledger

1. The approved mandate was committed and pushed to `main` at `3c440aa`.
2. The night branch was created from that exact anchor and its first Record checkpoint was committed at `8229b83`.
3. Draft pull request 52 is open. No experiment runner, smoke, pod, result, or cost exists; `exp069` spend remains 0 USD.

Visible messages added in CP-002

Assistant

2026-07-19 00:50:48.822 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate itself builds, but the initial activity-trace rendering found an unclosed Typst delimiter in a verbatim message containing bracket and currency punctuation. I'm escaping only that transcript presentation; the approved scientific contract remains unchanged.

Assistant

2026-07-19 00:52:07.816 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate is now preregistered on `main` at `3c440aa`; `exp069` spend is 0 USD and no compute has started. I'm creating `night/spiking-heidelberg-digits/ar070` from that exact anchor, adding the branch-start checkpoint, then pushing and opening the new draft PR.

Assistant

2026-07-19 00:53:19.217 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The new draft PR is open as 52 from the exact locked anchor. I'm adding its timestamped branch/PR checkpoint to `ar071` and pushing that final setup commit; `exp069` remains unrun with 0 USD spend and no active pods.

Checkpoint CP-003

This implementation-and-smoke checkpoint follows CP-002. Its immutable private source has SHA-256 prefix `8b2ba5ea34fb`, and its checkpoint time is `2026-07-19 01:39:56.429 UTC`. One private repository path in a visible handoff prompt is redacted; injected environment/context messages, hidden reasoning, and tool payloads are excluded.

Decision and action ledger

1. `Exp069` was implemented through `experiments/exp069.py` without editing `tools/snn`. The official-test route present in `exp068` was removed rather than inherited, and matched rasters were redirected to validation data.
2. Focused Ruff, type, and compilation checks passed.
3. The registered local smoke passed both cells. Split hashes match `exp068`; losses and rates were finite; skipped and non-finite update counters were zero; COBA E and PING E/I populations were active.
4. No cloud compute had started and spend remained 0 USD.

Visible messages added in CP-003

Assistant

2026-07-19 00:54:21.949 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Done.

- PR 51 merged into `main`: `569926f`

- Exp069 mandate locked on main: 3c440aa
- New branch: night/spiking-heidelberg-digits/ar070
- New draft PR: PR 52
- Activity checkpoint pushed: 08edd92
- Demolab build: 37 entries, zero stubs
- Current RunPod spend: \$0
- Active pods: 0
- No experiment or official-test access has started yet.

User

2026-07-19 01:14:49.989 UTC

Role: user · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

do it

User

2026-07-19 01:15:12.669 UTC

Role: user · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

give me the goal to give you so I can compact convo

Assistant

2026-07-19 01:15:30.248 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Use this after /compact:

/goal Continue and complete the locked exp069 autoresearch night on PR #52.

Repository and state:

- Work in (PRIVATE LOCAL PATH REDACTED).
- Current branch: night/spiking-heidelberg-digits/ar070.
- PR 51/exp068 was accepted and merged into main as 569926f.
- The exp069 mandate is locked in writings/ar070.typ on main at 3c440aa.
- Draft PR 52: <https://github.com/eoinmurray/pinglab/pull/52>
- Latest night checkpoint: 08edd92.
- Activity trace: writings/ar071.typ.
- Experiment/report IDs: exp069 and writings/exp069.typ.
- No exp069 compute has started; recorded spend is \$0 and there should be zero active pods.

Execute the registered design exactly:

- Test whether extending matched training from 40 to 80 epochs raises PING's selected validation accuracy by at least 3 percentage points over exp068's 43.50% baseline.
- One seed: 42.
- Reuse exp068's exact deterministic 7,340-training/816-validation split and verify its index hashes.
- Preserve every exp068 architecture, optimizer, input, readout, ordering, and preprocessing setting.
- The only scientific change is 80 training epochs instead of 40.
- COBA: inhibitory loop disabled and no voltage-gradient dampening.
- PING: inhibitory loop enabled and dampening 1000.

- Use the registered checkpoint rule: highest validation accuracy, then lower validation cross-entropy, then earlier epoch.
- The official SHD test is forbidden: do not stage, load, inspect, evaluate, or report it.
- No architecture-specific tuning, rescue, or post-result design changes.
- Do not edit tools/snn without asking first.

Execution protocol:

1. Run the registered local 128-training/128-validation, two-epoch smoke first.
2. Require finite optimization, active populations, matched partitions, and no persistent skipped updates.
3. If the gate passes, run one full 80-epoch cell per architecture.
4. RunPod authorization: at most two concurrent pods, one per architecture, with a hard total ceiling of \$10 and four-hour per-pod backstops.
5. Reap pods after completion or failure and verify zero remain.
6. Do not run the full test suite on the 4 GB host; use focused Ruff, ty, compilation/integrity checks, and demolab build.

Deliverables:

- Implement through experiments/exp069.py using existing CLI capabilities.
- Produce artifacts/data/exp069 with provenance, split hashes, numbers.json, reproducer, learning/activity curves, runtime and memory diagnostics, skipped/non-finite counters, and matched validation input/E/I rasters.
- Create writings/exp069.typ as a cold-readable artifact-driven report.
- Update the timestamped Record and Digest in writings/ar070.typ.
- Continue writings/ar071.typ with verbatim chronological user-visible messages and timestamped, hash-linked checkpoints. Exclude hidden reasoning and tool payloads; sanitize credentials, tokens, SSH material, IPs, private paths, environment values, pod identifiers, and sensitive infrastructure. Maintain immutable private raw transcript checkpoints outside the repository and record their SHA-256 prefix hashes.
- Add timestamped PR 52 comments at meaningful scientific and compute milestones.
- Make focused commits for implementation, smoke, dispatch, results, failures, and publication; preserve killed attempts.
- Build successfully, push the existing branch, and update PR 52
- Do not create another branch or PR, modify main, merge PR 52, or use the official test.
- Stop and ask before changing the registered scientific design, editing tools/snn, exceeding the compute budget, or taking another action requiring authority.
- Finish at the human publication gate with a concise evidence summary, exact compute spend, validation performed, rendered ar070/ar071/exp069 links, and PR 52 Leave it draft and unmerged.

Assistant

2026-07-19 01:33:02.681 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm resuming exp069 from the locked branch state. I'll first verify the Demolab runbook, worktree, PR/pod state, and mandate, then implement and smoke-test before any authorized compute dispatch. I'll also checkpoint this resumed visible thread into the sanitized activity log before publication.

Assistant

2026-07-19 01:33:14.646 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The branch is clean at checkpoint **08edd92**, tracks the intended remote, and the provider reports zero active pods, so the registered \$0 starting state is intact. I'm now reading the active mandate/runbook and comparing exp068's runner/artifacts so exp069 changes only the epoch count and test access is structurally excluded.

Assistant

2026-07-19 01:33:59.312 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The reuse audit found the key safety issue: exp068 contains a post-freeze official-test path, so exp069 cannot merely change **40** to **80**. I'm deriving the runner while deleting every download/evaluation route and changing rasters to fixed validation examples; the training recipe and selection logic remain identical.

Assistant

2026-07-19 01:35:43.759 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Focused Ruff, type, and compilation checks pass. The runner now pins the exact exp068 split hashes, sets only 80 epochs as the scientific change, uses a four-hour pod backstop, and has no official-test URL/import/evaluator. I'm starting the mandatory 128/128, two-epoch local smoke now; no cloud spend is involved.

Assistant

2026-07-19 01:39:56.429 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered smoke gate passed exactly: both cells were finite, all update counters stayed clean, COBA E activity was 23.93 Hz, and PING E/I activity was 2.95/14.56 Hz. The split hashes exactly match exp068. I'm checkpointing the implementation and smoke evidence—plus the resumed visible transcript—before dispatching either pod.

Checkpoint CP-004

This compute-dispatch checkpoint follows commit **b944ebf**. Its immutable private source has SHA-256 prefix **2006f03650d5**, and its checkpoint time is **2026-07-19 01:43:23.094 UTC**.

Provider pod identifiers and sensitive infrastructure details are excluded.

Compute ledger

1. Exactly two single-GPU pods were dispatched, one per architecture, at 0.99 USD/hour each. Four-hour backstops cap exposure at 7.92 USD.
2. A transient PING capacity miss was recovered by the existing bounded provisioning retry. The resulting active fleet was exactly two.
3. Both cells are pinned to **b944ebf**; official-test access remains forbidden.

Visible messages added in CP-004

Assistant

2026-07-19 01:42:35.132 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Smoke checkpoint **b944ebf** is pushed and the rendered trace/night PDFs were rebuilt. The gate authorizes full compute, so I'm posting the timestamped smoke milestone to PR 52, then I'll dry-run the exact two-cell dispatch before using `--live`. I'm applying the GitHub publishing workflow because these PR writes are explicitly part of the mandate.

Assistant

2026-07-19 01:42:46.629 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The dry-run resolves to exactly two 5090 pods—one COBA and one PING—with no extra jobs. The smoke checkpoint is pinned and the projected worst-case backstop cost remains below the authorized \$10 ceiling. I'm dispatching those two pods now; immediately afterward I'll record their creation time and verify the fleet count is exactly two.

Assistant

2026-07-19 01:43:23.094 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are now running at \$0.99/hour each, pinned to **b944ebf**; the first PING capacity request briefly found no stock, then the registered retry succeeded without creating an extra pod. Fleet count is exactly two, so the maximum four-hour exposure is \$7.92—inside the \$10 ceiling. I'm recording this dispatch milestone while the pods train.